

Human Augmentation for Skill Acquisition and Skill Transfer

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ABSTRACT

Human augmentation or augmented humans is regarded as an important research field with a view to the future society in which computer technology such as Virtual and Augmented Reality, Artificial Intelligence, Computer Vision, Robotics, etc. are highly integrated. Human augmentation is not just a research to make human much stronger. It should be used to assist people's daily activities. For example, it can be used to teach sports or musical performances more effectively. It can be used even for assisting people with disabilities. This workshop is expecting novel research results or late breaking results on designs, methods, implementations, or applications to augment or enhance human ability, in physical and intellectual, by using advanced technologies of VR/AR, AI, CV, and Robotics.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**; • **Human-centered computing** → **Human computer interaction (HCI)**; **Mixed / augmented reality**.

KEYWORDS

brain-machine interface; cyber training; human augmentation; musical training; rehabilitation; robotics; skill transfer; sports training; virtual and augmented reality;

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1 BACKGROUND

Human augmentation or augmented humans[9] is regarded as an important research field with a view to the future society in which computer technology such as Virtual and Augmented Reality[2, 8], Artificial Intelligence[4, 7], Robotics[5], etc. are highly integrated.

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One of the main motivations of human augmentation is to strengthen the people with a help of advanced technologies. There is a powered suit such as HAL by CYBERDYNE Inc. as a representative human augmentation technology. This is used to carry heavy objects that human cannot possess or to complement the lost body functions due to injuries and diseases.

On the other hand, there is another aspect in human augmentation which makes the human itself stronger by supporting he or she acquires special skills faster and more effectively. The advanced skills of top athletes, elite music performers[1], or people with disabilities[3, 6] are acquired by many years of training and experience. It is difficult to externalize such skills, and therefore it is difficult to transfer the skills to others. For example, in sports science, it is possible to measure body motion using the latest equipment such as special video equipment, small sensors, etc. However, the feedback to the athlete is limited to video playback after training and presentation of the measurement data. It is desirable that the coach intervenes in real time during training, and teach how to use the body, gaze direction, or psychological guidance etc.

In this workshop, we aim to discuss the technologies and case studies of skill acquisition support system that acquires (i.e. copies) advanced skills from people and transfers (i.e. pastes) them to others using computer vision, virtual and augmented reality, robotics, and artificial intelligence. In particular, we want to discuss the following 4 topics:

- **Advanced technologies for skill acquisition and transfer:**
First, we want to discuss the advanced technologies that copy the skills using computer vision, machine learning, and device technologies, and paste the skills using VR/AR or Robotics technologies.
- **Case studies in sports:**
Next, we want to discuss human augmentation systems in sports domain including advanced sports science, cyber training systems.
- **Case studies in music:**
Third, we want to discuss human augmentation systems in music domain including coaching systems for musical instruments.
- **Case studies in medicine:**
Finally, we want to discuss human augmentation systems in medicine including rehabilitation systems, surgery training systems.

2 ORGANIZERS

This workshop will be organised by an interdisciplinary team of researchers all of which are currently actively working in the field of Human Augmentation.

Hideki Koike is a Professor at Tokyo Institute of Technology. He has been working on information visualization of huge data, projector-camera systems, vision-based HCI such as interactive surfaces, and digital sports such as camera-embedded balls. He is currently leading a Japanese Government funded JST CREST project entitled "A Study on Skill Acquisition Mechanism and Development of Skill Transfer Systems".

Jun Rekimoto is a Professor at the University of Tokyo. His research interests include human-computer interaction, computer augmented environments and computer augmented human (human-computer integration). He invented various innovative interactive systems and sensing technologies. He also elected to ACM SIGCHI Academy.

Junichi Ushiba is an Associate Professor at the Department of Biosciences and Informatics, Faculty of Science and Technology, Keio University, Japan. Dr. Ushiba's research interests are focused on Brain-Machine Interface to guide brain plasticity as neurological rehabilitation. He takes computational, neuroimaging, and electrophysiological approaches to assess sensorimotor functions in the brains with medical quality.

Shinichi Furuya is a researcher at Sony Computer Science Laboratories Inc., where he leads the Music Excellence Project that aims at enhancing expertise of expert musicians by combining principles of the brain and body and technologies that assess and improve sensorimotor skills.

Asa Ito is an Associate Professor at Tokyo Institute of Technology. Research specialty is in aesthetics and contemporary art. She also has been studying people with disabilities and published many books including "How Blind People See the World", "Body Theory of Blind Athlete",

3 WEBSITE

Our website will be located at <https://sites.google.com/view/haa2021/> and will act as a portal both for this and future workshops.

4 PRE-WORKSHOP PLANS

We will distribute a CFP in all relevant communities, announcing the CFP on popular mailing lists and social media. We will also directly contact researchers and practitioners who are likely to be interested in the workshop and write to relevant institutions and research labs. Paper will be reviewed and selected using a juried process managed by the workshop organizers.

5 WORKSHOP STRUCTURE

The workshop is planned to last half day. There are 4 sessions where accepted authors will present their work in 5 minutes. In total, we are expecting 16 presentations.

5.1 Time Schedule

- 13:00: Brief Introduction & Welcome
- 13:10: Keynote Speaker + Q&A
- 13:30: Ten minutes break
- 13:40: Session 1: Technologies for Human Augmentation
- 14:20: Ten minutes break
- 14:30: Session 2: Human Augmentation in Sports
- 15:10: Ten minutes break
- 15:20: Session 3: Human Augmentation in Music
- 16:00: Ten minutes break
- 16:10: Session 4: Human Augmentation in Medicine
- 16:50: Closing

6 CALL FOR PARTICIPATION

We invite position papers for the CHI2021 Workshop on "Human Augmentation for Skill Acquisition and Skill Transfer". This half-day workshop will offer an interdisciplinary forum of discussion for both academics and industries.

Researchers and developers are invited to submit 4 pages of position paper in the CHI Extended Abstracts format (references excluded) of within the scope of challenges for Human Augmentation for Skill Acquisition and Skill Transfer. In particular, we are expecting, but not limited to, the following topics:

- Computer visions for acquiring skills
- Visual/audio/haptic feedback for transferring skills
- Deep Learning for skill abstraction
- Systems and case studies in sports training
- Systems and case studies in music training
- Systems and case studies in rehabilitation

All submissions are reviewed based on the relevance and the quality of the position paper by the workshop organizers. When accepted, each participant will upload his/her presentation video up to 5 minutes describing his/her work and will attend the session for Q&A and discussion. At least one author of each accepted paper must register for the workshop and all participants must register for the workshop and at least one day of the conference.

Please submit your paper via easychair system. You can find the link and details for the submission on our workshop website: <https://sites.google.com/view/haa2021/>

All accepted submissions to the workshop will be public on the website. Moreover, we plan to publish all the accepted papers using ACM International Conference Proceedings Series (ICPS).

Important dates:

- Submission deadline (Jan. 26, 2021)
- Notification date (Feb. 9, 2021)
- Camera ready due (Feb. 16, 2021)

7 POST-WORKSHOP PLANS

We intend to further collaborate with the participants of the workshop. Based on the discussions and networking established in the workshop, we plan to organize follow-up symposiums.

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